

Math 465: Introduction to Combinatorics

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Quizzes #1 and #2, as well as homeworks #1 and #2 have been graded.

Please DO NOT email me about regrades, Submit regrade requests through Gradescope. Please be respectful to your graders.

Homework #3 due tonight.

These slides will be posted on Canvas.

Linear recurrences: general set-up

We aim to solve **linear recurrences with constant coefficients**, i.e., recurrences of the form

$$(*) \quad h_n + a_1 h_{n-1} + a_2 h_{n-2} + \cdots + a_k h_{n-k} = 0.$$

Here

- k is the **order** of the recurrence;
- $a_1, \dots, a_k$ are its **coefficients**;
- $a_k \neq 0$ without loss of generality;
- the k **initial values** $h_0, h_1, \dots, h_{k-1}$ are given;
- n takes values $k, k+1, k+2, \dots$.

Example

$$h_n - 5h_{n-1} + 6h_{n-2} = 0$$

The theory of linear recurrences is very similar to the theory of linear differential equations.

The characteristic equation

$$(*) \quad h_n + a_1 h_{n-1} + a_2 h_{n-2} + \cdots + a_k h_{n-k} = 0$$

Lemma

For $q \neq 0$, the geometric progression $h_n = q^n$ satisfies $(*)$ if and only if

$$(**) \quad q^k + a_1 q^{k-1} + a_2 q^{k-2} + \cdots + a_k = 0.$$

Proof

$$\begin{aligned} q^n + a_1 q^{n-1} + a_2 q^{n-2} + \cdots + a_k q^{n-k} &= 0 \\ \iff q^k + a_1 q^{k-1} + a_2 q^{k-2} + \cdots + a_k &= 0 \end{aligned}$$

Equation $(**)$ is the **characteristic equation** for the recurrence $(*)$.

Example

$$(*) \quad h_n - 5h_{n-1} + 6h_{n-2} = 0$$

$$(**) \quad q^2 - 5q + 6 = 0$$

Roots: $q = 2$ and $q = 3$. Solutions $h_n = 2^n$ and $h_n = 3^n$.

The vector space of solutions of a linear recurrence

Lemma

The solutions of a linear recurrence

$$(*) \quad h_n + a_1 h_{n-1} + a_2 h_{n-2} + \cdots + a_k h_{n-k} = 0$$

form a vector space. The dimension of this vector space is k .

Example

$$(*) \quad h_n - 5h_{n-1} + 6h_{n-2} = 0$$

1	0	-6	-30	-114	...
0	1	5	19	65	...
a	b	$-6a+5b$	$-30a+19b$	$-114a+65b$	...

Geometric progressions

Lemma

Geometric progressions with distinct ratios are linearly independent.

Proof [sketch]

Suppose that k geometric progressions (in the example below, $k = 3$)

$$\begin{array}{cccc} 1 & q & q^2 & \cdots \\ 1 & r & r^2 & \cdots \\ 1 & s & s^2 & \cdots \end{array}$$

(with q, r, s distinct and nonzero) are linearly dependent. Then

$$\det \begin{pmatrix} 1 & q & q^2 \\ 1 & r & r^2 \\ 1 & s & s^2 \end{pmatrix} = 0 \Rightarrow \exists (a, b, c) \neq (0, 0, 0) \begin{cases} a + bq + cq^2 = 0, \\ a + br + cr^2 = 0, \\ a + bs + cs^2 = 0. \end{cases}$$

Thus $a + bt + ct^2$, a nonzero polynomial of degree $\leq k - 1$, has k distinct roots, a contradiction.

General solution in the case of distinct roots

$$(*) \quad h_n + a_1 h_{n-1} + a_2 h_{n-2} + \cdots + a_k h_{n-k} = 0$$

Three Lemmas

- The solutions of $(*)$ form a vector space of dimension k .
- A geometric progression $h_n = q^n$ satisfies $(*)$ if and only if

$$(**) \quad q^k + a_1 q^{k-1} + a_2 q^{k-2} + \cdots + a_k = 0.$$

- Geometric progressions with distinct ratios are linearly independent.

These lemmas imply the following theorem.

Theorem

If the characteristic equation $(**)$ has distinct roots $q_1, \dots, q_k$, then

$$h_n = \sum_{i=1}^k c_i q_i^n$$

is the general solution of $(*)$.

The case of distinct roots: example

Theorem

Given a linear recurrence

$$(*) \quad h_n + a_1 h_{n-1} + a_2 h_{n-2} + \cdots + a_k h_{n-k} = 0,$$

if the characteristic equation

$$q^k + a_1 q^{k-1} + a_2 q^{k-2} + \cdots + a_k = 0$$

has distinct roots $q_1, \dots, q_k$, then the general solution of $(*)$ is given by

$$h_n = \sum_{i=1}^k c_i q_i^n.$$

Example

The general solution of the recurrence $h_n - 5h_{n-1} + 6h_{n-2} = 0$ is given by $h_n = c_1 \cdot 2^n + c_2 \cdot 3^n$. For the initial conditions $h_0 = 1$ and $h_1 = -2$, we get $c_1 + c_2 = 1$ and $2c_1 + 3c_2 = -2$, implying $c_1 = 5$ and $c_2 = -4$.

Fibonacci numbers

Definition [Pingala, *ca.* 200 BCE; Leonardo di Pisa, 1202]

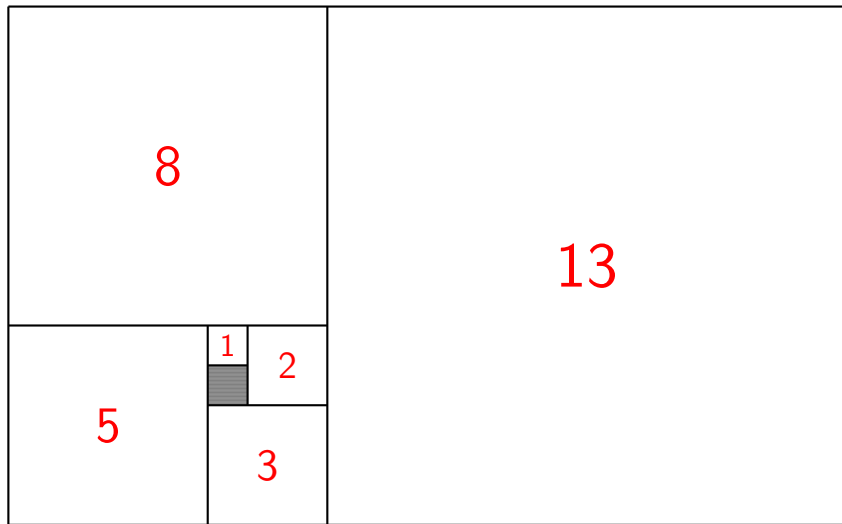
The **Fibonacci numbers** are defined by the recurrence

$$f_n = f_{n-1} + f_{n-2} \quad (n \geq 2)$$

together with the initial conditions $f_0 = 0$, $f_1 = 1$.

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
f_n	0	1	1	2	3	5	8	13	21	34	55	89	144	233	377

Fibonacci numbers and square tilings



Phyllotaxis



8 parallel rows of
scales spiralling
gradually



13 parallel rows of
scales spiralling at a
medium slope



21 parallel rows of
scales spiralling at a
steep slope

Spirals in a cactus



21 spiral rows counting in one direction and 34 in the other

Towards the formula for the Fibonacci numbers

The Fibonacci recurrence

$$f_n = f_{n-1} + f_{n-2}$$

has the characteristic equation

$$q^2 - q - 1 = 0.$$

The roots of this equation are

$$q_1 = \frac{1 + \sqrt{5}}{2}, \quad q_2 = \frac{1 - \sqrt{5}}{2}.$$

Consequently,

$$f_n = c_1 q_1^n + c_2 q_2^n,$$

for some constants c_1 and c_2 . To find c_1 and c_2 , we write

$$f_0 = 0 = c_1 + c_2,$$

$$f_1 = 1 = c_1 q_1 + c_2 q_2 = c_1(q_1 - q_2) = c_1 \sqrt{5}.$$

Thus $c_1 = \frac{1}{\sqrt{5}}$ and $c_2 = -\frac{1}{\sqrt{5}}$.

Formula for the Fibonacci numbers

Theorem

$$f_n = \frac{1}{\sqrt{5}} \left(\frac{1 + \sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1 - \sqrt{5}}{2} \right)^n.$$

Corollary

The Fibonacci number f_n is the closest integer to $\frac{1}{\sqrt{5}} \left(\frac{1 + \sqrt{5}}{2} \right)^n$.

Corollary

$$\lim_{n \rightarrow \infty} \frac{f_{n+1}}{f_n} = \frac{1 + \sqrt{5}}{2}.$$

Sums of 1's and 2's

Theorem

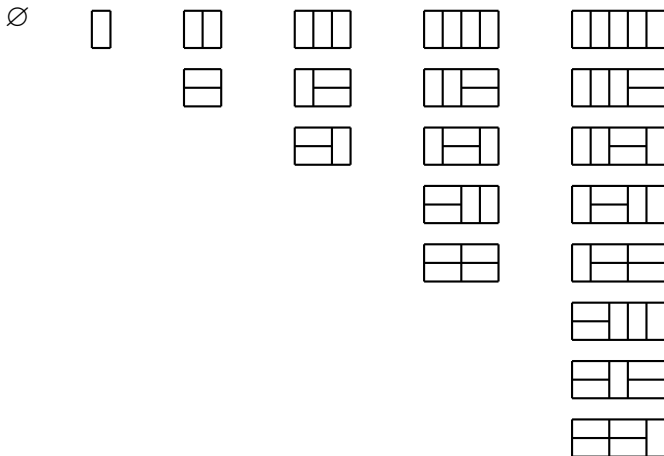
The number of ways to represent n as an ordered sum of summands each of which is equal to 1 or 2 is equal to the Fibonacci number f_{n+1} .

$\emptyset$	1	1+1	1+1+1	1+1+1+1	1+1+1+1+1
		2	1+2	1+1+2	1+1+1+2
			2+1	1+2+1	1+1+2+1
				2+1+1	1+2+1+1
				2+2	1+2+2
					2+1+1+1
					2+1+2
					2+2+1

Domino tilings

Theorem

The number of ways to tile a $2 \times n$ rectangular board by dominoes is equal to the Fibonacci number f_{n+1} .



Binary strings with no consecutive 1's

Theorem

The number of binary strings of length n that do not contain a pair of consecutive 1's is equal to the Fibonacci number f_{n+2} .

$\emptyset$	0	00	000	0000
	1	01	001	0001
		10	010	0010
			100	0100
			101	0101
				1000
				1001
				1010

Linear recurrences with constant coefficients, general case

Recall that we are interested in solving the linear recurrence

$$(*) \quad \sum_{i=0}^k a_i h_{n-i} = 0$$

where $n \geq k$, $a_0 = 1$ and $a_k \neq 0$.

We studied the case when the characteristic equation

$$(**) \quad \sum_{i=0}^k a_i q^{k-i} = 0$$

has distinct roots. What if it has repeated roots?

Roots of polynomials

Definition

We say that q_0 is a **root of multiplicity m** of a polynomial $P(q)$ (or of the equation $P(q) = 0$) if

$$P(q) = (q - q_0)^m R(q)$$

where $R(q)$ is a polynomial in q such that $R(q_0) \neq 0$.

Lemma [no proof here]

For a polynomial $P(q)$ and a number q_0 , the following are equivalent:

- q_0 is a root of $P(q)$ of multiplicity m ;
- q_0 is a root of $P(q), P'(q), P''(q), \dots, P^{(m-1)}(q)$ but not $P^{(m)}(q)$.

Fundamental Theorem of Algebra [no proof here]

Any polynomial $P(q)$ of degree $k > 0$ with complex coefficients has precisely k complex roots, counted with multiplicities.

New solutions for repeated roots (1)

Proposition

Let $q_0 \neq 0$ be a root of multiplicity ≥ 2 of the characteristic equation

$$(**) \quad \sum_{i=0}^k a_i q^{k-i} = 0.$$

Then $h_n = n q_0^n$ is a solution of the linear recurrence (*).

Proof

We want to show that

$$\sum_{i=0}^k a_i \cdot (n - i) q_0^{n-i} = 0.$$

Multiplying (**) by q^{n-k} , we see that q_0 is a repeated root of

$$\sum_{i=0}^k a_i q^{n-i} = 0.$$

Therefore q_0 is a root of the equation

$$\sum_{i=0}^k a_i \cdot (n - i) q^{n-i-1} = 0.$$

The claim follows.

New solutions for repeated roots (2)

Corollary

Let $q_0 \neq 0$ be a repeated root of the characteristic equation (**). Then both $h_n = q_0^n$ and $h_n = n q_0^n$ satisfy the linear recurrence (*).

Proposition/Exercise

Let q_0 be a root of multiplicity m of the characteristic equation (**). Then for any $\ell \in \{0, 1, \dots, m-1\}$, the sequence $h_n = n^\ell q_0^n$ satisfies (*).

As q_0 varies over all roots of (**), we obtain k such solutions, by the Fundamental Theorem of Algebra.

Proposition/Exercise

These k solutions are linearly independent. Consequently, they form a basis in the space of all solutions of (*).

General solution in the case of repeated roots

Theorem

Consider the linear recurrence

$$(*) \quad \sum_{i=0}^k a_i h_{n-i} = 0.$$

Let $q_1, \dots, q_r$ be the complex roots of its characteristic equation

$$(**) \quad \sum_{i=0}^k a_i q^{k-i} = 0,$$

with respective multiplicities $m_1, \dots, m_r$. Then the general solution of the recurrence $(*)$ is given by

$$h_n = C_1(n)q_1^n + \dots + C_r(n)q_r^n$$

where each $C_i(n)$ is a polynomial in n of degree $\deg(C_i) < m_i$.

Case of repeated roots: Example

Problem

Solve the linear recurrence $h_n = 3h_{n-2} + 2h_{n-3}$ with the initial conditions $h_0 = 0, h_1 = 0, h_2 = 1$.

n	0	1	2	3	4	5	6	7	8	9
h_n	0	0	1	0	3	2	9	12	31	54

Solution

The characteristic equation

$$q^3 - 3q - 2 = (q + 1)^2(q - 2) = 0$$

has roots $-1, -1, 2$. Hence $h_n = (c_1 + c_2 n)(-1)^n + c_3 \cdot 2^n$. Now

$$\begin{cases} c_1 + c_3 = 0 \\ -c_1 - c_2 + 2c_3 = 0 \\ c_1 + 2c_2 + 4c_3 = 1 \end{cases} \Rightarrow \begin{cases} c_1 = -\frac{1}{9} \\ c_2 = \frac{1}{3} \\ c_3 = \frac{1}{9} \end{cases} \Rightarrow h_n = \frac{(3n-1)(-1)^n + 2^n}{9}$$